

Communication (4)

1. For what value of k , if any, will $y = ke^{-2x} + 4\cos(3x)$ be a solution to the differential equation

$$\frac{d^2y}{dx^2} + 9y = 26e^{-2x}?$$

(2)

$$y = ke^{-2x} + 4\cos(3x)$$

$$\frac{dy}{dx} = -2ke^{-2x} - 12\sin(3x)$$

$$\frac{d^2y}{dx^2} = 4ke^{-2x} - 36\cos(3x)$$

$$4ke^{-2x} - 36\cos(3x) + 9(ke^{-2x} + 4\cos(3x)) = 26e^{-2x}$$

$$13ke^{-2x} = 26e^{-2x} \quad \checkmark \quad k=2 \quad \text{So } k \text{ must be equal to 2. will be a solution}$$

2. During a certain epidemic, the number of people that are infected at any time increases at a rate proportional to the number of people that are infected at that time. If 1,000 people are infected when the epidemic is first discovered, and 1,200 are infected 7 days later, how many people are infected 12 days after the epidemic is first discovered? (2)

$$\frac{dP}{dt} = \frac{d}{dt} \text{Rate proportional to population}$$

$$\int \frac{dP}{P} = \int k dt$$

$$\ln(P) = k \cdot t + C$$

$$P = Ce^{k \cdot t}$$

$$1000 = Ce^0 \quad \text{at time } t=0 \quad 1000 \text{ people are infected}$$

$$1200 = C$$

$$\text{At } t=7 \quad P=1200$$

$$P = 1000e^{k \cdot t}$$

$$= e^{k \cdot 7}$$

$$\frac{1200}{1000}$$

$$= k$$

$$\frac{\ln\left(\frac{12}{10}\right)}{7} = k \quad \text{Now plug in 12} \quad P = 1000e^{0.02604 \cdot 12 \text{ days}}$$

$$k = 0.02604 \quad P = 1366.8 \quad \text{more down}$$

$$P = 1367 \quad \text{So People infected is } 1366$$

Knowledge & Understanding

1. What is the general solution of the differential equation $\frac{dy}{dx} = \sqrt{y} - \sqrt{y} \sin x$ for $y > 0$? (2)

$$\frac{dy}{dx} = \sqrt{y}(1 - \sin x)$$

$$\int \frac{dy}{\sqrt{y}} = \int (1 - \sin x) dx \quad y = \left(\frac{x + \cos(x) + C}{2} \right)^2$$

2. Solve the initial value problem: $\frac{dy}{dx} = \frac{4\sqrt{y} \ln x}{x}$, $y = 1$ when $x = e$ (2)

$$\int \frac{dy}{\sqrt{y}} = 4 \int \frac{\ln(x)}{x} dx \quad y = \ln(|x|)^4$$

$$2\sqrt{y} = 4 \cdot \frac{\ln(x)^2}{2} + C$$

$$2\sqrt{y} = 2 \ln(x)^2 + C$$

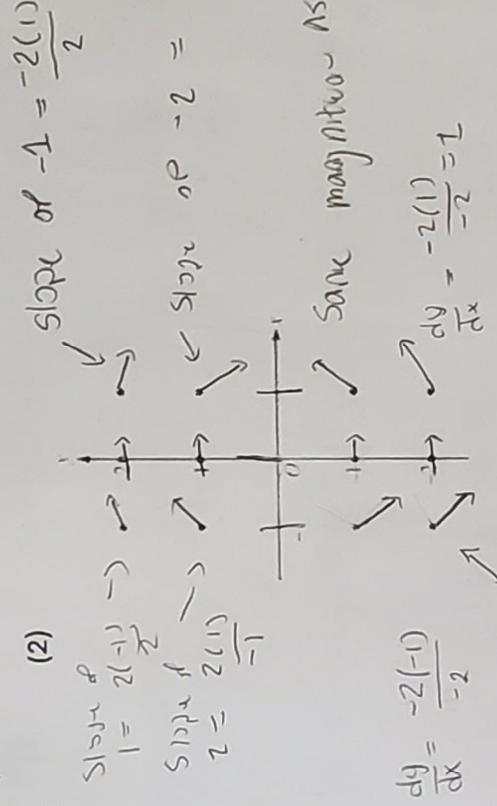
$$y = 1 \quad x = e$$

$$2\sqrt{y} = 4 \cdot \frac{y^2}{2} + C$$

Consider the differential equation $\frac{dy}{dx} = -\frac{2x}{y}$. In the grid provided, sketch a slope field for the 12 points indicated. Explain. (2)

Slope 0 when $y = 0$

Slope undefined when $y = 0$

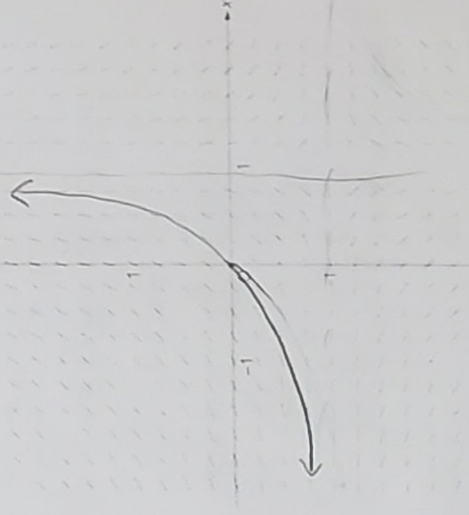


$$\frac{dy}{dx} = \frac{-2(-1)}{-2} = 1$$

Slope at $(-1, 1)$ as well

The slope field for a certain differential equation is given below. Which of the following could be a particular solution $y(0)=0$? Explain your selection. (2)

$$\frac{dy}{dx} = \frac{-1}{1-x}$$



$$\frac{dy}{dx} = \frac{1(-x) + x}{1-x} = \frac{1}{1-x}$$

$$(A) \quad y = \ln|1-x|$$

$$(B) \quad y = e^{-x^2} - 1$$

$$(C) \quad y = \frac{x}{1-x}$$

$$(D) \quad y = \frac{x^2}{1-x^2}$$

$$\frac{dy}{dx} = \frac{1}{1-x^2}$$

Not a solution as it has 2 asymptotes vertically, 1

$$y = (1-x^2)(2x) = x^2(-2x)$$

$$y = \frac{1}{(1-x)(1+x)}$$

5. Evaluate the following integrals:

a) $\int \frac{\left(\frac{1}{x^3}-4\right)^5}{6x^3} dx$

let $u = x^{1/3}-4$

$\frac{du}{dx} = \frac{1}{3}x^{-2/3}$

$3du = x^{-2/3} dx$

$\frac{1}{6} \int (x^{1/3}-4)^5 \cdot x^{2/3} dx$
 $\frac{1}{6} \int 3u^5 \cdot du = \frac{1}{12} (x^{1/3}-4)^6 + C$
 $\frac{1}{2} \int u^5 \cdot du = \frac{1}{2} \cdot \frac{u^6}{6} + C$

b) $\int \frac{17x+4}{3x^2-7x-6} dx$

Fraction decomposition

$\int \frac{17x+4}{(x-3)(3x+2)} dx$

$\frac{A}{x-3} + \frac{B}{3x+2} = \frac{17x+4}{(x-3)(3x+2)}$

$B=2$
 $A=5$

$\int \left(\frac{5}{x-3} + \frac{2}{3x+2} \right) dx$

$\left[5 \ln(x-3) + \frac{2}{3} \ln(3x+2) \right] + C$

$\frac{2}{3} \ln(16) - 5 \ln(3)$

$\ln(16^{2/3}) - \ln(3^5)$
 $\ln\left(\frac{16^{2/3}}{3^5}\right) \ln\left(\frac{2/3}{4/243}\right)$

c) $\int_{\pi/6}^{\pi/2} \sin^5(2x) \cos(2x) dx$

let $u = \sin(2x)$
 $\frac{du}{dx} = \cos(2x)$
 $\frac{1}{2} \int u^4 \cdot du = \frac{1}{2} \cdot \frac{u^5}{5} + C$
 $\frac{1}{10} \sin^5(2x) + C$

d) $\int e^x \sec(e^x) dx$

let $v = e^x$
 $\frac{dv}{dx} = e^x$
 $\int \sec(v) dv = \ln|\sec(v) + \tan(v)| + C$

$\ln|\sec(e^x) + \tan(e^x)| + C$

$\ln|\sec(e^x) + \tan(e^x)| + C$

$\ln|\sec(e^x) + \tan(e^x)| + C$

b) Use tabular integration: (2)

7. a) Use integration by parts to evaluate: (4)

a) $\int x^2 e^{-3x} dx$

$-\frac{1}{3} x^2 \cdot e^{-3x} + \frac{1}{3} \int e^{-3x} \cdot 2x dx$

$= -\frac{1}{3} x^2 \cdot e^{-3x} + \frac{1}{3} \left(-\frac{1}{3} e^{-3x} \cdot 2x + \frac{2}{3} \int e^{-3x} dx \right)$

$= -\frac{1}{3} x^2 \cdot e^{-3x} + \frac{1}{3} \left(-\frac{1}{3} e^{-3x} \cdot 2x + \frac{2}{3} \cdot \left(-\frac{1}{3} e^{-3x} \right) \right) + C$

$= -\frac{1}{3} x^2 \cdot e^{-3x} - \frac{2}{9} x \cdot e^{-3x} - \frac{2}{27} e^{-3x} + C$

$\int x^4 \sin(2x) dx$

$-4x^4 \cos(2x) + 12x^3 \sin(2x) - 24x^2 \cos(2x) + 24x \sin(2x) - 12 \cos(2x)$

$-12 \cos(2x)$

$= -\frac{1}{2} x^4 \cos(2x) + x^3 \sin(2x) + x^2 \cos(2x) - x \sin(2x) - \frac{3}{4} \cos(2x) + C$

numbers first

Application (8)

1. The rate at which a baby bird gains weight is proportional to the difference between its adult weight and its current weight. At time $t=0$, when the bird is first weighted, its weight is 20g. If $B(t)$ is the weight in grams, at time t days after it is first weighted then:

$$\frac{dB}{dt} = \frac{1}{5}(100 - B)$$

(3)

- a) Use separation of variables to determine the particular solution that satisfies the initial condition $B(0)=20$.

$$\frac{dB}{dt} = \frac{1}{5}(100 - B)$$

$$\int \frac{dB}{(100-B)} = \int \frac{1}{5} dt \quad \rightarrow 100 - 80 = 20$$

$$-\ln(100-B) = \frac{1}{5}t + C$$

$$100-B = Ce^{-1/5t}$$

$$100-C = 20$$

$$80 = C$$

2. A wild animal preserve can support no more than 250 gorillas. Twenty-eight gorillas were known to be in the preserve in 1970. Assuming that the rate of growth of the population is:

$$\frac{dP}{dt} = 0.0004P(250 - P), \text{ where time } t \text{ is in years.}$$

(4)

- a) Using partial fractions technique find a formula for the gorilla population in terms of t .

$$\int \frac{dP}{P(250-P)} = \int 0.0004 dt$$

$$\frac{A}{P} + \frac{B}{250-P} =$$

$$\frac{1}{250} \left(\int \frac{1}{P} dP + \int \frac{1}{250-P} dP \right) = 0.0004t + C$$

$$\ln(P) - \ln(250-P) = 0.1t + C$$

$$\ln\left(\frac{250-P}{P}\right) = -0.1t + C$$

$$\frac{250-P}{P} = Ce^{-0.1t}$$

$$\frac{250}{P} - 1 = Ce^{-0.1t}$$

$$\frac{250}{P} = 1 + Ce^{-0.1t}$$

$$P = \frac{250}{1 + Ce^{-0.1t}}$$

$$P = \frac{250}{1 + 7.9285e^{-0.1t}}$$

$$A(250-P), B(P) = 1$$

$$B = \frac{1}{1+7.9285e^{-0.1t}}$$

$$P = \frac{250}{1 + 7.9285e^{-0.1t}}$$

$$1 + 7.9285e^{-0.1t} = \frac{250}{P}$$

$$e^{-0.1t} = \frac{250}{P} - 1$$

$$t = \frac{\ln\left(\frac{250}{P} - 1\right)}{-0.1}$$

$t = 55.12$ years to reach carrying capacity (250 gorillas)